

BASH SCRIPTING ASSIGNMENT 2

1. Create a bash script to **check if a directory is available or not.**

```
#!/bin/bash
```

```
read -p "Enter directory path: " DIR
```

```
if [ -d "$DIR" ]; then
```

```
    echo "Directory exists: $DIR"
```

```
else
```

```
    echo "Directory does not exist: $DIR"
```

```
fi
```

Run it

```
chmod +x check_directory.sh
```

```
./check_directory.sh
```

Example:

Enter directory path: /var/log

Directory exists: /var/log

```
#!/bin/bash

read -p "Enter directory path: " DIR

if [ -d "$DIR" ]; then
    echo "Directory exists: $DIR"
else
    echo "Directory does not exist: $DIR"
fi

[ec2-user@ip-172-31-14-118 ~]$ ./check_directory.sh
Enter directory path: /var/log
Directory exists: /var/log
[ec2-user@ip-172-31-14-118 ~]$
```

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2. Create a bash script to **create multiple files**.

```
#!/bin/bash
```

```
read -p "Enter directory where files should be created: " DIR
```

```
read -p "Enter number of files to create: " COUNT
```

```
mkdir -p "$DIR"
```

```
for ((i=1; i<=COUNT; i++))
```

```
do
```

```
    touch "$DIR/file$i.txt"
```

```
done
```

```
echo "$COUNT files created successfully in $DIR"
```

To Run

```
chmod +x create_files.sh
```

```
./create_files.sh
```

```
[ec2-user@ip-172-31-14-118 ~]$ vi create_file.sh
[ec2-user@ip-172-31-14-118 ~]$ chmod +x create_files.sh
chmod: cannot access 'create_files.sh': No such file or directory
[ec2-user@ip-172-31-14-118 ~]$ chmod +x create_file.sh
[ec2-user@ip-172-31-14-118 ~]$ ./create_file.sh
Enter directory where files should be created: /tmp/test
Enter number of files to create: 5
5 files created successfully in /tmp/test
[ec2-user@ip-172-31-14-118 ~]$ |
```

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```
#!/bin/bash

read -p "Enter directory where files should be created: " DIR
read -p "Enter number of files to create: " COUNT

mkdir -p "$DIR"

for ((i=1; i<=COUNT; i++))
do
    touch "$DIR/file$i.txt"
done

echo "$COUNT files created successfully in $DIR"
~

[ec2-user@ip-172-31-14-118 ~]$ chmod +x create_file.sh
[ec2-user@ip-172-31-14-118 ~]$ ./create_file.sh
Enter directory where files should be created: /tmp/test
Enter number of files to create: 5
5 files created successfully in /tmp/test
[ec2-user@ip-172-31-14-118 ~]$ ls -l /tmp/test
total 0
-rw-r--r--. 1 ec2-user ec2-user 0 Aug 23 11:15 file1.txt
-rw-r--r--. 1 ec2-user ec2-user 0 Aug 23 11:15 file2.txt
-rw-r--r--. 1 ec2-user ec2-user 0 Aug 23 11:15 file3.txt
-rw-r--r--. 1 ec2-user ec2-user 0 Aug 23 11:15 file4.txt
-rw-r--r--. 1 ec2-user ec2-user 0 Aug 23 11:15 file5.txt
[ec2-user@ip-172-31-14-118 ~]$ vi create_file.sh
[ec2-user@ip-172-31-14-118 ~]$ |
```

3. Create a bash script totake a **backup of a directory**.

```
#!/bin/bash
```

```
SOURCE_DIR="/var/www/html"
```

```
BACKUP_DIR="/backup"
```

```
DATE=$(date +%Y-%m-%d_%H-%M-%S)
```

```
mkdir -p "$BACKUP_DIR"
```

```
tar -czf "$BACKUP_DIR/html_backup_$(DATE).tar.gz" "$SOURCE_DIR"
```

```
if [ $? -eq 0 ]; then
```

```
    echo "Backup completed successfully."
```

```
    echo "Backup file: $BACKUP_DIR/html_backup_$(DATE).tar.gz"
```

```
else
```

```
    echo "Backup failed."
```

```
fi
```

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To Run

```
chmod +x backup_directory.sh
```

```
sudo ./backup_directory.sh
```

```
#!/bin/bash

SOURCE_DIR="/var/www/html"
BACKUP_DIR="/backup"

DATE=$(date +%Y-%m-%d_%H-%M-%S)

mkdir -p "$BACKUP_DIR"

tar -czf "$BACKUP_DIR/html_backup_$DATE.tar.gz" "$SOURCE_DIR"

if [ $? -eq 0 ]; then
    echo "Backup completed successfully."
    echo "Backup file: $BACKUP_DIR/html_backup_$DATE.tar.gz"
else
    echo "Backup failed."
fi
```

```
[ec2-user@ip-172-31-14-118 ~]$ sudo ./backup_directory.sh
tar: Removing leading `/' from member names
Backup completed successfully.
Backup file: /backup/html_backup_2026-08-23_11-29-11.tar.gz
[ec2-user@ip-172-31-14-118 ~]$ |
```

4. Create a bash script to **install Nginx on an EC2 server**.

```
#!/bin/bash
```

```
echo "Updating packages..."
```

```
sudo dnf update -y
```

```
echo "Installing Nginx..."
```

```
sudo dnf install nginx -y
```

```
echo "Starting Nginx..."
```

```
sudo systemctl start nginx
```

```
echo "Enabling Nginx at boot..."
```

```
sudo systemctl enable nginx
```

```
echo "Checking Nginx status..."
```

```
sudo systemctl status nginx --no-pager
```

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```
echo "Nginx installation completed."
```

To Run

```
chmod +x install_nginx.sh
```

```
sudo ./install_nginx.sh
```

```
[ec2-user@ip-172-31-14-118 ~]$  
[ec2-user@ip-172-31-14-118 ~]$ vi install_nginx.sh  
[ec2-user@ip-172-31-14-118 ~]$ chmod +x install_nginx.sh  
[ec2-user@ip-172-31-14-118 ~]$ nginx  
-bash: nginx: command not found  
[ec2-user@ip-172-31-14-118 ~]$ ./install_nginx.sh  
Updating packages...  
Last metadata expiration check: 0:05:15 ago on Sun Aug 23 11:28:56 2026.  
Dependencies resolved.  
Nothing to do.  
Complete!  
Installing Nginx...  
Last metadata expiration check: 0:05:16 ago on Sun Aug 23 11:28:56 2026.  
Dependencies resolved.  
=====
```

Package	Architecture	Version
Installing:		
nginx	x86_64	1:1.30.4-1.amzn2023.0.1
Installing dependencies:		
generic-logos-httpd	noarch	18.0.0-12.amzn2023.0.3
gperftools-libs	x86_64	2.9.1-1.amzn2023.0.3
libunwind	x86_64	1.4.0-5.amzn2023.0.3
nginx-core	x86_64	1:1.30.4-1.amzn2023.0.1
nginx-filesystem	noarch	1:1.30.4-1.amzn2023.0.1
nginx-mimetypes	noarch	2.1.49-3.amzn2023.0.3

```
#!/bin/bash
```

```
echo "Updating packages..."
```

```
sudo dnf update -y
```

```
echo "Installing Nginx..."
```

```
sudo dnf install nginx -y
```

```
echo "Starting Nginx..."
```

```
sudo systemctl start nginx
```

```
echo "Enabling Nginx at boot..."
```

```
sudo systemctl enable nginx
```

```
echo "Checking Nginx status..."
```

```
sudo systemctl status nginx --no-pager
```

```
echo "Nginx installation completed."|
```

```
~
```

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5. Create a bash script to install **Apache Tomcat** on an EC2 server.

```
#!/bin/bash

set -e

echo "===== "

echo "  TOMCAT INSTALLATION STARTING"

echo "===== "

TOMCAT_VERSION="10.1.59"

TOMCAT_DIR="/opt/tomcat"

TOMCAT_USER="tomcat"

TOMCAT_GROUP="tomcat"

echo

echo "[1/10] Checking root privileges..."

if [ "$(id -u)" -ne 0 ]; then

    echo "ERROR: Run this script as root."

    echo "Use: sudo -i"

    exit 1

fi

echo "Running as root."

echo

echo "[2/10] Installing required packages..."

dnf install -y java-17-amazon-corretto wget tar

echo

echo "[3/10] Checking Java..."

java -version

echo

echo "[4/10] Removing incomplete previous Tomcat installation..."
```

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```
systemctl stop tomcat 2>/dev/null || true

rm -rf "$TOMCAT_DIR"

echo

echo "[5/10] Creating Tomcat user and group..."

if ! getent group "$TOMCAT_GROUP" >/dev/null; then

    groupadd "$TOMCAT_GROUP"

fi

if ! id "$TOMCAT_USER" >/dev/null 2>&1; then

    useradd \

        --system \

        --gid "$TOMCAT_GROUP" \

        --home-dir "$TOMCAT_DIR" \

        --shell /sbin/nologin \

        "$TOMCAT_USER"

fi

echo

echo "[6/10] Creating Tomcat directory..."

mkdir -p "$TOMCAT_DIR"

cd /tmp

echo

echo "[7/10] Downloading Apache Tomcat ${TOMCAT_VERSION}..."

TOMCAT_FILE="apache-tomcat-${TOMCAT_VERSION}.tar.gz"

TOMCAT_URL="https://dlcdn.apache.org/tomcat/tomcat-10/v${TOMCAT_VERSION}/bin/${TOMCAT_FILE}"

wget -O "$TOMCAT_FILE" "$TOMCAT_URL"

if [ ! -f "$TOMCAT_FILE" ]; then
```

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```
echo "ERROR: Tomcat download failed."

exit 1

fi

echo

echo "Download successful."

echo

echo "[8/10] Extracting Tomcat..."

tar -xzf "$TOMCAT_FILE" \

-C "$TOMCAT_DIR" \

--strip-components=1

echo

echo "[9/10] Setting permissions..."

chown -R "$TOMCAT_USER:$TOMCAT_GROUP" "$TOMCAT_DIR"

find "$TOMCAT_DIR/bin" -type f -name "*.sh" -exec chmod 755 {} \;

chmod 755 "$TOMCAT_DIR"

echo

echo "Checking Tomcat files..."

ls -la "$TOMCAT_DIR"

echo

echo "Checking Tomcat executable scripts..."

ls -la "$TOMCAT_DIR/bin/*.sh"

echo

echo "[10/10] Starting Tomcat..."

sudo -u "$TOMCAT_USER" "$TOMCAT_DIR/bin/startup.sh"

echo
```

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```
echo "=====
```

```
echo "    TOMCAT INSTALLATION COMPLETE"
```

```
echo "=====
```

```
echo
```

```
echo "Checking Tomcat process..."
```

```
ps -ef | grep '[t]omcat'
```

```
echo
```

```
echo "Checking port 8080..."
```

```
ss -lntp | grep ':8080' || true
```

```
echo
```

```
echo "Tomcat URL:"
```

```
echo "http://YOUR_EC2_PUBLIC_IP:8080"
```

To Run

```
chmod +x install_tomcat.sh
```

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```
sudo ./install_tomcat.sh
```

```
#!/bin/bash

set -e

echo "=====
echo "          TOMCAT INSTALLATION STARTING
echo "=====

TOMCAT_VERSION="10.1.59"
TOMCAT_DIR="/opt/tomcat"
TOMCAT_USER="tomcat"
TOMCAT_GROUP="tomcat"

echo
echo "[1/10] Checking root privileges..."

if [ "$(id -u)" -ne 0 ]; then
    echo "ERROR: Run this script as root."
    echo "Use: sudo -i"
    exit 1
fi

echo "Running as root."

echo
echo "[2/10] Installing required packages..."

dnf install -y java-17-amazon-corretto wget tar

echo
echo "[3/10] Checking Java..."

java -version

echo
echo "[4/10] Removing incomplete previous Tomcat installation..."
```

```
[root@ip-172-31-14-118 ec2-user]# vi install_tomcat.sh
[root@ip-172-31-14-118 ec2-user]# rm -rf install_tomcat.sh
[root@ip-172-31-14-118 ec2-user]# vi install_tomcat.sh
[root@ip-172-31-14-118 ec2-user]# chmod +x install_tomcat.sh
[root@ip-172-31-14-118 ec2-user]# ./install_tomcat.sh
=====
TOMCAT INSTALLATION STARTING
=====

[1/10] Checking root privileges...
Running as root.

[2/10] Installing required packages...
Last metadata expiration check: 0:26:06 ago on Sun Aug 23 11:28:56 2026.
Package java-17-amazon-corretto-1:17.0.20+8-1.amzn2023.1.x86_64 is already installed.
Package wget-1.21.3-1.amzn2023.0.5.x86_64 is already installed.
Package tar-2:1.34-1.amzn2023.0.4.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!

[3/10] Checking Java...
openjdk version "17.0.20" 2026-07-21 LTS
OpenJDK Runtime Environment Corretto-17.0.20.8.1 (build 17.0.20+8-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.20.8.1 (build 17.0.20+8-LTS, mixed mode, sharing)

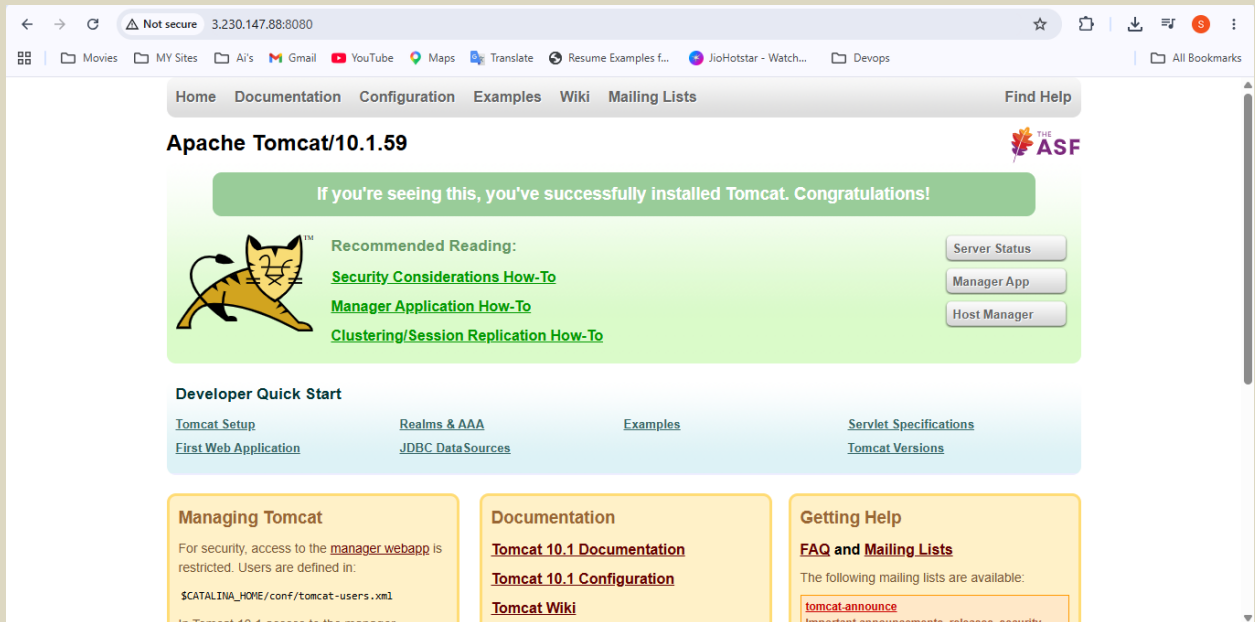
[4/10] Removing incomplete previous Tomcat installation...

[5/10] Creating Tomcat user and group...

[6/10] Creating Tomcat directory...

[7/10] Downloading Apache Tomcat 10.1.59...
--2026-08-23 11:55:03-- https://d1cdn.apache.org/tomcat/tomcat-10/v10.1.59/bin/apache-tomcat-10.1.59.tar.gz
Resolving d1cdn.apache.org (d1cdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to d1cdn.apache.org (d1cdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14424246 (14M) [application/x-gzip]
```

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6. Create a bash script to **check if the Nginx service is running**, if not running then script should start the service.

```
#!/bin/bash
```

```
if systemctl is-active --quiet nginx
```

```
then
```

```
    echo "Nginx is running."
```

```
else
```

```
    echo "Nginx is NOT running."
```

```
    echo "Starting Nginx..."
```

```
    sudo systemctl start nginx
```

```
    if systemctl is-active --quiet nginx
```

```
    then
```

```
        echo "Nginx started successfully."
```

```
    else
```

```
        echo "Failed to start Nginx."
```

```
fi
```

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fi

To Run

chmod +x check_nginx.sh

sudo ./check_nginx.sh

```
#!/bin/bash

if systemctl is-active --quiet nginx
then
    echo "Nginx is running."
else
    echo "Nginx is NOT running."
    echo "Starting Nginx..."

    sudo systemctl start nginx

    if systemctl is-active --quiet nginx
    then
        echo "Nginx started successfully."
    else
        echo "Failed to start Nginx."
    fi
fi
```

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```
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[root@ip-172-31-14-118 ec2-user]# ./check_nginx.sh
Nginx is running.
[root@ip-172-31-14-118 ec2-user]# |
```

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.

To engage with the community please visit community.nginx.org.

For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

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7. Create a bash script for a **calculator**.

```
#!/bin/bash

echo "===== "

echo "  BASH CALCULATOR"

echo "===== "

read -p "Enter first number: " NUM1

read -p "Enter operator (+ - * / %): " OP

read -p "Enter second number: " NUM2

case "$OP" in

  +)

    RESULT=$((NUM1 + NUM2))

    ;;

  -)

    RESULT=$((NUM1 - NUM2))

    ;;

  \*)

    RESULT=$((NUM1 * NUM2))

    ;;

  /)

    if [ "$NUM2" -eq 0 ]; then

      echo "Error: Cannot divide by zero."

      exit 1

    fi

    RESULT=$((NUM1 / NUM2))

    ;;

  %)
```

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```
if [ "$NUM2" -eq 0 ]; then  
    echo "Error: Cannot divide by zero."  
    exit 1  
fi  
  
RESULT=$((NUM1 % NUM2))  
  
;;  
  
*)  
    echo "Invalid operator."  
    exit 1  
    ;;  
  
esac  
  
echo "Result = $RESULT"
```

To Run

```
chmod +x calculator.sh
```

```
./calculator.sh
```

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```
[root@ip-172-31-14-118 ec2-user]# vi calculator.sh
[root@ip-172-31-14-118 ec2-user]# chmod +x calculator.sh
[root@ip-172-31-14-118 ec2-user]# ./calculator.sh
```

```
=====
      BASH CALCULATOR
=====
Enter first number: 3
Enter operator (+ - * / %): +
Enter second number: 3
Result = 6
[root@ip-172-31-14-118 ec2-user]# |
```

```
#!/bin/bash
```

```
echo "=====
      BASH CALCULATOR
=====

read -p "Enter first number: " NUM1
read -p "Enter operator (+ - * / %): " OP
read -p "Enter second number: " NUM2

case "$OP" in
    +)
        RESULT=$((NUM1 + NUM2))
        ;;
    -)
        RESULT=$((NUM1 - NUM2))
        ;;
    \*)
        RESULT=$((NUM1 * NUM2))
        ;;
    /)
        if [ "$NUM2" -eq 0 ]; then
            echo "Error: Cannot divide by zero."
            exit 1
        fi
        RESULT=$((NUM1 / NUM2))
        ;;
    %)
        if [ "$NUM2" -eq 0 ]; then
            echo "Error: Cannot divide by zero."
            exit 1
```

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8. Create a bash script to **check if a directory exists**, if not then create a directory.

```
#!/bin/bash

read -p "Enter directory path: " DIR

if [ -d "$DIR" ]

then

    echo "Directory already exists: $DIR"

else

    echo "Directory does not exist."

    echo "Creating directory..."

    mkdir -p "$DIR"

    if [ $? -eq 0 ]

    then

        echo "Directory created successfully: $DIR"

    else

        echo "Failed to create directory."

    fi

fi
```

To Run

```
chmod +x create_directory.sh
./create_directory.sh
```

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```
#!/bin/bash

read -p "Enter directory path: " DIR

if [ -d "$DIR" ]
then
    echo "Directory already exists: $DIR"
else
    echo "Directory does not exist."
    echo "Creating directory..."

    mkdir -p "$DIR"

    if [ $? -eq 0 ]
    then
        echo "Directory created successfully: $DIR"
    else
        echo "Failed to create directory."
    fi
fi

~
~
~

[root@ip-172-31-14-118 ec2-user]# vi create_directory.sh
[root@ip-172-31-14-118 ec2-user]# ls
create_directory.sh
[root@ip-172-31-14-118 ec2-user]# chmod +x create_directory.sh
[root@ip-172-31-14-118 ec2-user]# ls
create_directory.sh
[root@ip-172-31-14-118 ec2-user]# ./create_directory.sh
Enter directory path: mkdir -p /tmp/devops/project
Directory does not exist.
Creating directory...
Directory created successfully: mkdir -p /tmp/devops/project
[root@ip-172-31-14-118 ec2-user]# |
```

9. Create a bash script to **delete the last 3 lines of a file.**

```
#!/bin/bash

read -p "Enter filename: " FILE

if [ ! -f "$FILE" ]

then

    echo "File does not exist."

    exit 1

fi

LINES=$(wc -l < "$FILE")

if [ "$LINES" -lt 3 ]

then
```

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```
    echo "File contains less than 3 lines."

    exit 1

fi

head -n -3 "$FILE" > "${FILE}.tmp"

mv "${FILE}.tmp" "$FILE"

echo "Last 3 lines deleted successfully."
```

To Run

```
chmod +x delete_last_3.sh
```

```
./delete_last_3.sh
```

```
#!/bin/bash

read -p "Enter filename: " FILE

if [ ! -f "$FILE" ]
then
    echo "File does not exist."
    exit 1
fi

LINES=$(wc -l < "$FILE")

if [ "$LINES" -lt 3 ]
then
    echo "File contains less than 3 lines."
    exit 1
fi

head -n -3 "$FILE" > "${FILE}.tmp"

mv "${FILE}.tmp" "$FILE"

echo "Last 3 lines deleted successfully."
~
```

```
[ec2-user@ip-172-31-14-118 ~]$ ls
create_directory.sh  delete_last_3.sh  file.txt  'mkdir -p '
[ec2-user@ip-172-31-14-118 ~]$ vi file.txt
[ec2-user@ip-172-31-14-118 ~]$ ./delete_last_3.sh
Enter filename: file.txt
Last 3 lines deleted successfully.
[ec2-user@ip-172-31-14-118 ~]$ |
```

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10. Bash script to monitor cpu and if it is more than 80% then send email notification.

```
#!/bin/bash

THRESHOLD=80

EMAIL="your-email@example.com"

CPU_USAGE=$(top -bn1 | grep "Cpu(s)" | \
awk '{print 100 - $8}' | cut -d. -f1)

echo "Current CPU usage: $CPU_USAGE%"

if [ "$CPU_USAGE" -gt "$THRESHOLD" ]

then

    SUBJECT="ALERT: High CPU Usage on $(hostname)"

    MESSAGE="CPU usage on $(hostname) is currently ${CPU_USAGE}%.

    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"

    echo "CPU alert email sent."

else

    echo "CPU usage is normal."

fi

sudo dnf install mailx -y
```

To Run

```
chmod +x cpu_monitor.sh
./cpu_monitor.sh
```

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```
#!/bin/bash

THRESHOLD=80
EMAIL="your-email@example.com"

CPU_USAGE=$(top -bn1 | grep "Cpu(s)" | \
awk '{print 100 - $8}' | cut -d. -f1)

echo "Current CPU usage: $CPU_USAGE%"

if [ "$CPU_USAGE" -gt "$THRESHOLD" ]
then
    SUBJECT="ALERT: High CPU Usage on $(hostname)"

    MESSAGE="CPU usage on $(hostname) is currently ${CPU_USAGE}%."

    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"

    echo "CPU alert email sent."
else
    echo "CPU usage is normal."
fi

cpu_monitor.sh create_directory.sh delete_last_3.sh file.txt 'mkdir -p '
[ec2-user@ip-172-31-14-118 ~]$ chmod +x cpu_monitor.sh
[ec2-user@ip-172-31-14-118 ~]$ ls
cpu_monitor.sh create_directory.sh delete_last_3.sh file.txt 'mkdir -p '
[ec2-user@ip-172-31-14-118 ~]$ ./cpu_monitor.sh
Current CPU usage: 3%
CPU usage is normal.
[ec2-user@ip-172-31-14-118 ~]$ |
```

11. Bash script to monitor disk space and if it is more than 80% then send email notification.

```
#!/bin/bash

THRESHOLD=80

EMAIL="your-email@example.com"

DISK_USAGE=$(df / | awk 'NR==2 {print $5}' | tr -d '%')

echo "Current disk usage: $DISK_USAGE%"

if [ "$DISK_USAGE" -gt "$THRESHOLD" ]
then

    SUBJECT="ALERT: High Disk Usage on $(hostname)"

    MESSAGE="Disk usage on $(hostname) is currently ${DISK_USAGE}%."

    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"

    echo "Disk alert email sent."
```

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```
else
```

```
    echo "Disk usage is normal."
```

```
fi
```

To Run

```
chmod +x disk_monitor.sh
```

```
./disk_monitor.sh
```

```
#!/bin/bash

THRESHOLD=80
EMAIL="your-email@example.com"

DISK_USAGE=$(df / | awk 'NR==2 {print $5}' | tr -d '%')

echo "Current disk usage: $DISK_USAGE%"

if [ "$DISK_USAGE" -gt "$THRESHOLD" ]
then
    SUBJECT="ALERT: High Disk Usage on $(hostname)"
    MESSAGE="Disk usage on $(hostname) is currently ${DISK_USAGE}%."
    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"
    echo "Disk alert email sent."
else
    echo "Disk usage is normal."
fi
```

```
cpd_monitor.sh  create_directory.sh  delete_files.sh
[ec2-user@ip-172-31-14-118 ~]$ chmod +x disk_monitor.sh
[ec2-user@ip-172-31-14-118 ~]$ ./disk_monitor.sh
Current disk usage: 25%
Disk usage is normal.
```

BASH SCRIPTING ASSIGNMENT 2

Bash script to monitor memory and if it is more than 80% then send email notification.

```
#!/bin/bash
```

```
THRESHOLD=80
```

```
EMAIL="your-email@example.com"
```

```
MEMORY_USAGE=$(free | awk '/Mem:/ {
```

```
    printf "%.0f", ($3/$2)*100
```

```
  }')
```

```
echo "Current memory usage: $MEMORY_USAGE%"
```

```
if [ "$MEMORY_USAGE" -gt "$THRESHOLD" ]
```

```
then
```

```
    SUBJECT="ALERT: High Memory Usage on $(hostname)"
```

```
    MESSAGE="Memory usage on $(hostname) is currently ${MEMORY_USAGE}%."
```

```
    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"
```

```
    echo "Memory alert email sent."
```

```
else
```

```
    echo "Memory usage is normal."
```

```
fi
```

To Run

```
chmod +x memory_monitor.sh
```

```
./memory_monitor.sh
```

BASH SCRIPTING ASSIGNMENT 2

```
#!/bin/bash

THRESHOLD=80
EMAIL="your-email@example.com"

MEMORY_USAGE=$(free | awk '/Mem:/ {
    printf "%.0f", ($3/$2)*100
}')

echo "Current memory usage: $MEMORY_USAGE%"

if [ "$MEMORY_USAGE" -gt "$THRESHOLD" ]
then
    SUBJECT="ALERT: High Memory Usage on $(hostname)"

    MESSAGE="Memory usage on $(hostname) is currently ${MEMORY_USAGE}%."

    echo "$MESSAGE" | mail -s "$SUBJECT" "$EMAIL"

    echo "Memory alert email sent."
else
    echo "Memory usage is normal."
fi|
~
~

[ec2-user@ip-172-31-14-118 ~]$ ./memory_monitor.sh
Current memory usage: 33%
Memory usage is normal.
[ec2-user@ip-172-31-14-118 ~]$ |
```

BASH SCRIPTING ASSIGNMENT 2

Exercise: Crontab Entries

April 5th Midnight

5th of Every November, January, June **if it is a Thursday**

At 05 and 27th minutes of 9, 10, 11 hours every day

34 minutes of 9th hour on 15th August

Every midnight

Every Weekend (**Saturday night 11:59 PM**)

After every reboot

Script:

```
#!/bin/bash
```

```
if [ "$(date +%u)" -eq 4 ]
```

```
then
```

```
    echo "Today is the 5th and it is Thursday."
```

```
    # Your command here
```

```
fi
```

Complete Crontab:

```
# 1. April 5th at midnight
```

```
0 0 5 4 * /path/to/script.sh
```

```
# 2. 5th of January, June and November
```

```
# Script itself must verify that the 5th is Thursday
```

```
0 0 5 1,6,11 * /path/to/check_thursday.sh
```

```
# 3. At 05 and 27 minutes of 9, 10 and 11 hours every day
```

```
5,27 9,10,11 * * * /path/to/script.sh
```

BASH SCRIPTING ASSIGNMENT 2

4. 34 minutes of 9th hour on 15th August

```
34 9 15 8 * /path/to/script.sh
```

5. Every midnight

```
0 0 * * * /path/to/script.sh
```

6. Every Saturday at 11:59 PM

```
59 23 * * 6 /path/to/script.sh
```

7. After every reboot

```
@reboot /path/to/script.sh
```

How to Configure Crontab:

```
sudo dnf install -y cronie
```

```
sudo systemctl enable --now crond
```

```
systemctl status crond
```

```
crontab -e
```

```
#!/bin/bash
```

```
echo "Cron executed at $(date)" >> /home/ec2-user/cron_output.log
```

```
~
```

```
[ec2-user@ip-172-31-14-118 ~]$ ls
```

```
mycron.sh
```

```
[ec2-user@ip-172-31-14-118 ~]$ chmod +x mycron.sh
```

```
[ec2-user@ip-172-31-14-118 ~]$ ls
```

```
mycron.sh
```

```
[ec2-user@ip-172-31-14-118 ~]$ cat /home/ec2-user/cron_output.log
```

```
cat: /home/ec2-user/cron_output.log: No such file or directory
```

```
[ec2-user@ip-172-31-14-118 ~]$ ./mycron.sh
```

```
[ec2-user@ip-172-31-14-118 ~]$ cat /home/ec2-user/cron_output.log
```

```
Cron executed at Sun Aug 23 13:36:23 UTC 2026
```

```
[ec2-user@ip-172-31-14-118 ~]$ |
```